

## EISENSTADT, Austria

WMO: 111900

Lat: 47.854N

Lon: 16.538E

Elev: 597

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 15.5       | 19.2 | 7.4                           | 8.2 | 17.4 | 12.3 | 10.5 | 22.6 | 23.3                  | 41.9 | 21.4 | 40.0 | 6.1                   | 340  | 0.513 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.6                   | 89.4            | 70.8 | 86.0 | 69.3 | 82.8 | 67.8 | 72.1                | 85.9 | 70.5 | 83.3 | 69.0 | 80.5 | 8.4                  | 120  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 67.5                            | 103.4 | 78.4 | 66.0 | 97.8 | 76.8 | 64.5 | 92.9 | 75.1 | 36.2          | 86.2 | 34.7 | 83.2 | 33.4 | 80.7 | 77.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |       |            |       |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |       | n=50 years |       |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min   | Max        |       |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |
| 21.1              | 18.7 | 16.5 | DB                         | 11.0 | 94.1 | 5.2                | 4.1                                                | 7.3       | 97.1 | 4.3        | 99.5 | 1.4        | 101.9 | -2.4       | 104.9 |
|                   |      |      | WB                         | 10.1 | 74.7 | 4.9                | 1.5                                                | 6.5       | 75.8 | 3.6        | 76.7 | 0.9        | 77.5  | -2.7       | 78.6  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 52.3   | 32.8  | 36.2  | 43.7  | 53.2  | 60.6  | 67.3  | 70.9  | 70.2  | 61.5  | 52.1  | 43.1  | 34.5 |
|                                                                                  | DBStd       | 15.13  | 8.01  | 8.16  | 7.77  | 7.27  | 6.30  | 6.27  | 5.87  | 5.83  | 6.08  | 7.01  | 7.11  | 7.31 |
|                                                                                  | HDD50       | 1970   | 535   | 390   | 223   | 51    | 3     | 0     | 0     | 0     | 1     | 61    | 226   | 480  |
|                                                                                  | HDD65       | 5215   | 999   | 808   | 661   | 358   | 165   | 48    | 15    | 18    | 139   | 402   | 658   | 944  |
|                                                                                  | CDD50       | 2791   | 1     | 2     | 26    | 146   | 331   | 520   | 649   | 627   | 345   | 125   | 18    | 1    |
|                                                                                  | CDD65       | 561    | 0     | 0     | 0     | 3     | 28    | 117   | 199   | 180   | 33    | 1     | 0     | 0    |
|                                                                                  | CDH74       | 4328   | 0     | 0     | 0     | 33    | 224   | 917   | 1607  | 1354  | 187   | 6     | 0     | 0    |
|                                                                                  | CDH80       | 1354   | 0     | 0     | 0     | 3     | 32    | 266   | 555   | 467   | 31    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.6    | 7.4   | 7.7   | 7.9   | 8.2   | 8.3   | 7.6   | 7.9   | 7.3   | 7.5   | 7.0   | 7.1   | 7.1  |
| Precipitation                                                                    | PrecAvg     | 24.5   | 1.1   | 1.0   | 1.6   | 1.6   | 2.6   | 3.0   | 3.0   | 2.8   | 2.6   | 1.9   | 1.8   | 1.4  |
|                                                                                  | PrecMax     | 32.4   | 2.5   | 2.6   | 3.5   | 3.5   | 6.9   | 7.3   | 6.9   | 6.5   | 6.7   | 3.8   | 3.7   | 3.0  |
|                                                                                  | PrecMin     | 16.9   | 0.2   | 0.0   | 0.4   | 0.0   | 0.5   | 0.8   | 0.2   | 0.3   | 0.7   | 0.1   | 0.0   | 0.2  |
|                                                                                  | PrecStd     | 3.9    | 0.6   | 0.7   | 0.9   | 0.9   | 1.4   | 1.6   | 1.7   | 1.5   | 1.6   | 1.0   | 0.9   | 0.7  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 56.7  | 59.9  | 68.6  | 78.2  | 83.8  | 90.9  | 93.4  | 94.3  | 83.9  | 74.0  | 65.2  | 55.8 |
|                                                                                  |             | MCWB   | 47.0  | 49.1  | 53.9  | 61.0  | 66.7  | 71.7  | 72.2  | 70.0  | 66.8  | 61.8  | 56.1  | 48.0 |
|                                                                                  | 2%          | DB     | 51.5  | 54.8  | 63.8  | 72.9  | 79.5  | 86.7  | 89.4  | 89.2  | 79.1  | 69.1  | 60.5  | 50.6 |
|                                                                                  |             | MCWB   | 44.5  | 46.4  | 51.6  | 58.2  | 64.7  | 70.2  | 70.7  | 70.6  | 65.0  | 59.5  | 53.2  | 44.8 |
|                                                                                  | 5%          | DB     | 47.5  | 50.9  | 59.1  | 68.9  | 76.3  | 82.8  | 86.0  | 85.1  | 75.4  | 65.6  | 56.2  | 46.8 |
|                                                                                  |             | MCWB   | 41.9  | 44.2  | 49.2  | 56.1  | 63.2  | 68.2  | 69.2  | 68.9  | 63.0  | 57.9  | 50.7  | 41.8 |
|                                                                                  | 10%         | DB     | 44.1  | 47.3  | 55.2  | 65.4  | 72.7  | 79.5  | 82.6  | 81.5  | 71.8  | 62.1  | 52.9  | 43.9 |
|                                                                                  |             | MCWB   | 39.6  | 41.9  | 47.3  | 54.2  | 60.9  | 66.4  | 67.9  | 67.2  | 61.6  | 55.9  | 48.5  | 39.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 48.3  | 50.1  | 54.9  | 62.3  | 68.3  | 73.9  | 74.1  | 73.7  | 68.2  | 62.9  | 57.0  | 48.8 |
|                                                                                  |             | MCDB   | 54.8  | 59.3  | 66.8  | 75.5  | 80.5  | 86.8  | 88.7  | 87.9  | 80.0  | 71.2  | 63.6  | 54.1 |
|                                                                                  | 2%          | WB     | 45.4  | 47.1  | 52.3  | 59.2  | 66.0  | 71.4  | 72.2  | 71.5  | 66.2  | 60.7  | 53.9  | 45.4 |
|                                                                                  |             | MCDB   | 51.0  | 53.7  | 61.7  | 70.9  | 77.6  | 83.9  | 86.3  | 85.6  | 76.4  | 67.6  | 59.2  | 50.3 |
|                                                                                  | 5%          | WB     | 42.3  | 44.7  | 50.1  | 57.1  | 63.9  | 69.4  | 70.5  | 69.8  | 64.7  | 58.7  | 51.3  | 42.5 |
|                                                                                  |             | MCDB   | 46.4  | 50.2  | 57.7  | 67.1  | 74.5  | 80.8  | 83.3  | 83.0  | 72.9  | 64.0  | 55.7  | 46.3 |
|                                                                                  | 10%         | WB     | 40.0  | 42.3  | 48.1  | 55.0  | 62.0  | 67.6  | 68.9  | 68.2  | 62.9  | 56.8  | 48.9  | 40.4 |
|                                                                                  |             | MCDB   | 43.8  | 46.9  | 54.5  | 63.9  | 70.8  | 77.6  | 80.2  | 79.1  | 69.9  | 61.4  | 52.3  | 43.3 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.8   | 11.1  | 14.0  | 16.6  | 16.7  | 17.1  | 17.6  | 17.3  | 14.8  | 12.9  | 9.4   | 7.9  |
|                                                                                  |             | MCDBR  | 14.0  | 17.2  | 20.8  | 22.2  | 22.5  | 22.6  | 23.0  | 23.2  | 20.4  | 17.5  | 14.8  | 13.3 |
|                                                                                  |             | MCWBR  | 9.2   | 10.6  | 11.6  | 10.9  | 10.3  | 9.4   | 8.7   | 8.8   | 8.7   | 9.4   | 9.2   | 9.1  |
|                                                                                  | 5% WB       | MCDBR  | 13.3  | 16.2  | 19.4  | 21.1  | 21.3  | 21.8  | 21.7  | 21.8  | 19.0  | 16.1  | 13.8  | 12.8 |
|                                                                                  |             | MCWBR  | 9.4   | 10.6  | 11.3  | 10.7  | 10.1  | 9.5   | 8.9   | 9.0   | 8.8   | 9.1   | 8.9   | 9.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.317  | 0.342 | 0.380 | 0.424 | 0.422 | 0.427 | 0.433 | 0.433 | 0.406 | 0.376 | 0.348 | 0.316 |      |
|                                                                                  | taud        | 2.391  | 2.352 | 2.273 | 2.177 | 2.229 | 2.254 | 2.238 | 2.250 | 2.297 | 2.354 | 2.389 | 2.417 |      |
|                                                                                  | Ebn at Noon | 239    | 257   | 263   | 262   | 267   | 266   | 262   | 256   | 252   | 240   | 223   | 224   |      |
|                                                                                  | Edh at Noon | 22     | 28    | 36    | 43    | 43    | 42    | 42    | 40    | 35    | 28    | 22    | 19    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 325    | 583   | 962   | 1433  | 1697  | 1847  | 1793  | 1596  | 1141  | 705   | 368   | 252   |      |
|                                                                                  | RadStd      | 35     | 79    | 121   | 140   | 175   | 132   | 153   | 154   | 135   | 75    | 35    | 29    |      |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            |  | +1.28 | N/A     | N/A    | +1.93   | N/A   | N/A   | -490        | -542  | N/A   | N/A   |
| Regional (84 neighbors) |  | +1.03 | N/A     | N/A    | +1.58   | +0.86 | +0.74 | -214        | -286  | +164  | +88   |

Nomenclature: See separate page